Voice Assistant — | المساعد الصوتي Specification v1.0

Prepared for Mohamed Gamal — 7 Nov 2026 — Save this file in the repository root as VOICE.md.

> ملاحظة بالعربي: الملف ده مستقل ومش تكملة لـ CHANGES.md. بيوصف ميزة واحدة بس: مساعد صوتي جوه التطبيق بيشتغل بقواعد محلية، من غير ذكاء اصطناعي ومن غير إنترنت. مكتوب بالإنجليزي عشان الوكيل (Antigravity) يفهمه بدقة، وجمل الأوامر والأسئلة بالعربي المصري لأنها اللي هتتقال فعلاً. لو أي حاجة مش واضحة، اسألني في الشات.

0. Instructions to the AI coding agent (read first)

1. This file is standalone. SPEC.md and CHANGES.md describe the app that already exists (its modules and services). This file adds ONE feature. If this file conflicts with them about the voice assistant, this file wins. About everything else, they win. Any earlier idea of an AI or cloud assistant has been dropped: there is none.
2. The owner is NOT a programmer. Talk to him in simple Egyptian Arabic. Give every action he must take as short numbered steps.
3. Work in the steps of section 12, one step at a time. After each step: run all tests, update README.md, CHANGELOG.md and DECISIONS.md, commit, tag (voice-1, voice-2...), let CI build the APK, give a plain-Arabic test checklist for that step, and wait for approval.
4. Do not add anything outside this file. If something is unclear, or you want a different approach, stop and ask first with 2 or 3 options and a recommendation.
5. Never wipe or reset user data. Any database change is a NEW Drift migration with a test.
6. No INTERNET permission, no networking library, no AI or LLM call, no analytics. The only external pieces are the Android platform speech recognizer and the Android text-to-speech engine, used through platform APIs.
7. Reuse the existing services and repositories. Every write made by voice goes through the same code as the manual screens (status rules, reminder scheduling, events, soft delete). Do not copy business logic.
8. Be honest about what depends on the phone (speech recognition accuracy, offline behavior, Arabic voice for speech output on Xiaomi). Measure it, report the numbers, and never claim a result you did not measure.
9. Every new APK must install over the previous one (same signing key, higher version).

1. What the owner wants (agreed)

- Speak commands to record things instead of typing: tasks, errands, reminders, expenses, income, habit logs, notes, ideas, wishlist items, health items. Also speak status changes: done, postponed with a reason, waiting, stopped, partial.
- Ask questions and get answers from his own data: what is ahead today, tomorrow, in two days or on a weekday; what he did yesterday, last week, this month or on a given date; spending questions; habit questions; what is overdue, postponed, waiting.
- It does not have to be AI. The idea must work, locally and predictably.
- The app will be published to other people later. This local design needs no API key, has no per-user cost, and sends no data anywhere, so it is also ready for publishing.

Decision	Choice
Starting and stopping	Tap to start listening, tap again to stop
Confirmation card	Confirmed by tap OR by voice (say 'تمام' or 'إلغاء')
Replies to questions	Text card plus spoken reply
Commands per sentence	One
Deleting by voice	Not available
Languages	Egyptian Arabic first, English later as an option
Internet	Not used by the app

2. The experience (what the owner sees and hears)

2.1 Entry points. A microphone button in the Today top bar, an item in More, and an Android app shortcut (long-press the app icon) named 'سجّل بالصوت / Voice'. All open the same screen: the assistant sheet.

2.2 The assistant sheet. A large microphone button, a live transcript area, a text field (so the same parser also accepts typed sentences), suggestion chips ('عملت', 'ورايا إيه النهارده؟', 'سجّل مصروف', 'إيه امبارح؟'), and the result area.

2.3 Flow.

1. Tap the microphone: a short vibration, a sound wave, and the transcript appears live.
2. Tap again to stop (maximum 60 seconds).
3. Understanding is instant and local. The result is one of three things: a Confirmation card (a command), an Answer card (a question), or 'Not understood'.

2.4 Confirmation card. Shows the interpreted action in plain words with every field, for example: مهمة — 'أكلم العميل' — الأحد 8 نوفمبر — 5:00 م — منبه قبلها 10 دقايق. Buttons: تأكيد / Confirm, تعديل / Edit (opens the normal pre-filled quick-add sheet), إلغاء / Cancel. After Confirm, the normal Undo snackbar appears.

- Voice confirmation: when a card appears, the microphone listens for a short window (default 5 seconds, a visible indicator shows it). The words تمام، أكد، تأكيد، ماشي، أيوه confirm. The words إلغاء، لأ، لا، سيبك cancel. The word تعديل opens the edit sheet. Anything else is ignored. Voice confirmation applies to the latest card only and can be switched off in Settings.
- Dates and times are always shown in words on the card, and amounts in digits with the currency, because speech recognition is weakest on numbers.

2.5 Answer card. A title (for example 'النهارده — السبت 7 نوفمبر'), sections with icons and counts, and buttons that open the matching screen (the day or period in the timeline, the finance screen, and so on). The same answer is spoken with text-to-speech when spoken replies are on.

2.6 Missing or ambiguous information.

- No date spoken for a task: the card shows the chips اليوم / بكرة / بدون تاريخ / اختر تاريخ, with 'بدون تاريخ' selected. One tap changes it.
- A name matches several tasks (for example two called التقرير): the assistant asks which one, with choices.
- A reason is required but missing (postponed, waiting, stopped, partial): the assistant asks once, with the owner's reason chips.
- A numeric date such as 12/9/2026 is read as day/month/year (Egypt) and shown in words ('12 سبتمبر 2026'), so he can catch a mistake.

2.7 Not understood. The assistant says what it heard ('سمعت: ...') and offers 'افتح الإضافة بالنص ده', which opens quick add with the text already in the title. The sentence is saved to a local list of unrecognized sentences (section 8) so patterns can be added later.

2.8 Limits (say so in the app help). It is not a free chat. It does not remember the conversation and does not answer open questions such as 'ليه صرفت كتير؟'. It handles one command per sentence. It cannot delete anything.

3. Speech input and speech output

3.1 Recognition. Use the Android SpeechRecognizer through a maintained Flutter plugin or a small platform channel. Locale ar-EG, falling back to ar. Partial results on. The permission is RECORD_AUDIO, which the app already has.

3.2 Tap to start, tap to stop. The Android recognizer ends a session by itself after some silence. The app must keep listening until the owner taps stop: chain sessions, merge the results, avoid duplicated words and lost words at session boundaries, and stop at 60 seconds. Test this carefully on the owner's phone.

3.3 What depends on the phone. Accuracy for the Egyptian dialect, whether recognition works in airplane mode, and whether audio is sent to Google services all depend on the device and its speech packages. Do not assume. Run the test of section 10.3 and report it. If recognition does not work offline, show a clear message with a shortcut to the system speech settings, and do not hide it. The typed-sentence field always works.

3.4 Speech output. Use the Android text-to-speech engine with an Arabic voice. Settings: on or off (default ON, as the owner chose), speed, and a tap to stop speaking. If no Arabic voice is installed, fall back to text only and show a short guide to install one. Answers about finance and health are spoken only if the owner turns on 'قراءة الردود الحساسة صوتياً' (default OFF), because a spoken answer can be overheard. Speak at most 5 list items, then say 'وكمان N'. Do not speak amounts when amounts are masked (the eye icon of the finance module).

3.5 Audio is not stored. The assistant never saves the recognized audio. The only exception is the explicit commands 'سجّل ملاحظة صوتية' and 'سجّل فكرة صوتية', which use the existing voice-notes feature.

4. Understanding sentences (the rule-based parser)

4.1 Pipeline. Normalize the text, detect the command or question pattern, extract slots (date, time, amount, quantity, category, reason, title), build an Intent with the missing slots, then the ActionPlanner turns it into a Confirmation card or an Answer card. The parser sits behind an IntentParser interface. This version has one implementation, RuleBasedIntentParser. Patterns, synonyms and phrase variants live in data files (JSON assets), so adding a variant does not need new code.

4.2 Normalization (matching only; never change what the owner sees). Remove diacritics and punctuation, unify alef forms (إ أ آ → ا), ة → ه, ى → ي, convert Arabic-Indic digits to Western digits, convert Arabic number words to numbers, expand common dialect variants from a dictionary (for example النهارده, النهاردة, انهارده).

4.3 Numbers in words. A pure Dart converter for Egyptian number words: واحد، اتنين، تلاتة، أربعة، خمسة، ستة، سبعة، تمانية، تسعة، عشرة، حداشر، اتناشر، تلتاشر، أربعتاشر، خمستاشر، عشرين، تلاتين، أربعين، خمسين، مية، ميتين، تلتمية، ألف، ألفين، تلات تلاف، and combinations such as 'ألف وخمسمية' and 'ونص'. Money words: جنيه، جنية، ج, EGP, قرش. Default currency EGP. Always show the result as digits on the card.

4.4 Dates and times (one shared DateTimeResolver, pure Dart, unit-tested with a fixed 'now').

- Day words: بعد بكرة، بعد النهارده، امبارح، أول امبارح, N أيام / يوم / يومين (بعد يومين = 2+). بعد أسبوع.
- Weekday names (السبت، الأحد، الاتنين، التلات، الأربع، الخميس، الجمعة): the next upcoming date, today excluded. With 'الجاي' or 'القادم' it is still the next upcoming one. With 'اللي فات' or 'الماضي' it is the most recent past one.
- Periods (used by questions): الأسبوع ده، الأسبوع اللي فات، الأسبوع الجاي، الشهر ده، الشهر اللي فات، السنة دي. The week starts on the day chosen in Settings (default Saturday).
- Explicit dates: '12/9', '12/9/2026', '12 سبتمبر', 'يوم 12 سبتمبر 2026'. Numeric dates are day/month/year. A missing year means the current year. Month names: يناير، فبراير، مارس، أبريل، مايو، يونيو، يوليو، أغسطس، سبتمبر، أكتوبر، نوفمبر، ديسمبر.
- A period word in a command (for example 'الأسبوع الجاي') does not define one date, so the card asks with date chips.
- A date in the past in a command shows a visible 'التاريخ فات' chip.
- Times: '8' الصبح, '9' العصر, '5' إلا تلت, 'ربع' إلا ربع, 'ربع ونص خمسة الساعة', '5:30' الساعة 5. Markers: الصبح / صباحاً = AM; بالليل / مساءً = الضهر / بعد الضهر / العصر = 12:00; بالليل' = الضهر = PM. With no marker: 1 to 6 = PM, 7 to 11 = AM, 12 = noon. The card shows AM or PM clearly and the owner can edit.
- 'قبلها بساعة' or 'قبلها بنص ساعة' sets the reminder offset.

Expected results with the fixed now = Saturday 7 Nov 2026 10:00 (device time zone Africa/Cairo, week starts Saturday):

- النهارده → 07-11-2026; بكرة → 08-11-2026 (Sunday); بعد بكرة → 09-11-2026; بعد يومين → 09-11-2026; بعد 3 أيام → 10-11-2026.
- امبارح → 06-11-2026; أول امبارح → 05-11-2026.
- الخميس → 12-11-2026; السبت → 14-11-2026 (today excluded).
- يوم 12/9 → 12-09-2026 (day/month).
- الأسبوع ده → 07-11-2026 to 13-11-2026; الأسبوع اللي فات → 31-10-2026 to 06-11-2026.
- الشهر ده → 01-11-2026 to 30-11-2026; الشهر اللي فات → 01-10-2026 to 31-10-2026.

5. Commands (what the owner can say)

5.1 Rules for all commands.

- One command per sentence. A command word is optional when the sentence has a date and an intent verb. For example the owner's own style 'انا بكرة عايز اعمل مونتاج الاعلان' must create a task.

- The title is the rest of the sentence after removing the command word, the date phrase, the time phrase and filler words (انا، عايز، هـ، لازم، اني). The title keeps the original spelling from the transcript, not the normalized spelling. If the title is empty, the assistant asks 'تسجل إيه؟'.
- Cards use the app's existing defaults (default reminder type and offsets from the Settings of Round 1).

5.2 Commands. Every command ends in a Confirmation card and then runs through the existing service named in the last column.

Intent	Examples (spoken)	Card and action
create_task	'سجّل بكرة أكلم العميل الساعة 5' / 'انا بكرة عايز اعمل مونتاج الاعلان' / 'يوم الخميس هقابل الدكتور' / 'ضيف مهمة اراجع الفاتورة'	Task with title, date, time, reminder from defaults. Same code as quick add.
create_errand	'سجّل مشوار بكرة اجيب البنزين' / 'مشوار النهارده البنك الساعة 12'	Errand. The word مشوار is required. Location is the text after من or عند, if spoken.
create_reminder	'فكرني بكرة الساعة 9 اكلم احمد' / 'نبهني بعد يومين ادفع الفاتورة'	Task with a reminder at the spoken time (default type from Settings).
update_status	'خلصت مهمة التقرير' / 'ابتديت المونتاج' / 'اجّل مهمة التقرير لبكرة لان مفيش فلوس' / 'مهمة العميل مستنية رد الدكتور' / 'وقفت مهمة الاعلان بسبب الميزانية' / 'عملت نص مهمة التصميم لاني تعبت'	Status values: done, in progress, postponed (date and reason), waiting (reason), stopped (reason), partial (progress and reason). Same status service, rules and events as the status sheet.
log_expense	'سجّل شاي بعشرة جنيه' / 'صرفت 50 على اكل بره' / 'دفعت 300 فاتورة النت امبارح'	Expense with category, amount, essential or side, date. Finance service.
log_income	'دخل 5000 من عميل' / 'قبضت 8000 مرتب'	Income.
log_habit	'شربت قهوة' / 'كوبايتين قهوة' / 'صليت الفجر' / 'قريت 5 صفحات قران' / 'عملت اذكار الصباح'	Habit log: +1, a quantity, or a checklist tick. Habits service.
create_note	'ملاحظة اللي بيذاكر بيتعلم الصبر' / 'سجّل ملاحظة ...'	Note. The words حكمة or اقتباس make its kind quote.

Intent	Examples (spoken)	Card and action
create_idea	'فكرة تطبيق للمشاوير' / 'سجّل فكرة ...'	Idea.
record_voice_note	'سجّل فكرة صوتية' / 'سجّل ملاحظة صوتية'	Opens the recorder and saves a voice-only idea or note (existing voice-notes feature). No transcription.
create_wishlist_item	'ضيف للمشتريات سماعة بألف وخمسمية' / 'عايز اشتري كاميرا'	Wishlist item with an optional price.
create_health_item	'فكرني اروح كشف الأسنان الخميس' / 'محتاج اعمل تحليل سكر'	Health item with status needed and an optional date.

A command for a module that does not exist yet, or that is turned off, answers with a clear message and does nothing.

5.3 Matching rules.

- Task lookup (update_status): compare normalized title tokens and compute a score. Prefer tasks due today, then overdue, then other open ones. Two or more tasks within 10 percent of the best score: show choices. No match: say 'مالقيتش مهمة اسمها ...' and offer to create it.
- Expense category: 1) the labels of the quick favorites, 2) category names, 3) the synonyms file (شاي، قهوة، كافيه، اكل بره، مطعم، بقالة، بنزين، مواصلات، فاتورة، نت، تليفون، دوا، علاج ...), 4) otherwise 'أخرى' with category chips on the card.
- Habit: by name, built-in key and synonyms. 'كوباية' alone is ambiguous between coffee and water, so ask with chips.
- Reason: by the owner's reasons list and synonyms (مفيش فلوس، معييش فلوس، الميزانية). Unknown text becomes a custom reason.
- Never guess silently. Ask, or show the chip.

6. Questions (what the owner can ask)

Each answer is built by the same services as the dashboards, so numbers always match the screens.

Intent	Examples (spoken)	Answer card
agenda	'ورايا ايه النهارده؟' / 'ورايا ايه بعد يومين؟' / 'عندي ايه يوم الخميس؟' / 'ورايا ايه الاسبوع ده؟'	Tasks and errands in time order, overdue count, obligations due, health appointments, and (for today only) habits not done yet.
recap	'عملت ايه امبارح؟' / 'الاسبوع اللي فات عملت ايه؟' / 'ايه اللي حصل يوم 12/9؟' / 'ملخص الشهر ده'	Done count, postponed count with the top reason, habit completion, spending total, notes and ideas counts, highlights, and a button to open the day or period in the timeline.
spending	'صرفت كام على اكل بره الشهر ده؟' / 'اكتر حاجة بصرف عليها؟' / 'مصروف النهارده كام؟'	Totals, top categories, essential vs side, comparison with the previous period.
habit_status	'شربت كام قهوة النهارده؟' / 'صليت ايه النهارده؟' / 'التزامي بالقران الاسبوع ده؟'	Progress and streak.
pending	'ايه المتأخر عندي؟' / 'ايه المؤجل؟' / 'ايه اللي مستنني؟'	Lists by status with their reasons.
obligations	'عليا ايه الشهر ده؟' / 'ايه الالتزامات المتأخرة؟'	Late and upcoming obligations.
health_pending	'ايه الكشوفات المؤجلة؟'	Needed and postponed items and how long each has waited.
wishlist	'اجمالي المشتريات اللي عايزها كام؟'	Wishlist totals.

Rules:

- When a question is about a date, the answer starts with the date in words ('12 يوم سبتمبر 2026').
- A question about a module that does not exist yet or is off gets a clear message.
- Before the timeline of Round 7 exists, the recap of a single day shows only what each module can answer (tasks, expenses, habits, notes).
- Spoken text comes from a separate SpeechComposer with Arabic templates and a plural helper: ١ مهمة، ٢ مهمتين، ٣ إلى ١٠ مهام، ١١ فأكثر مهمة. The helper is unit-tested.

7. Execution, safety and privacy

- Flow: parser → Intent → ActionPlanner → Confirmation card → owner confirms → the existing service executes → Undo snackbar.
- Not available by voice: deleting anything, permanent delete, emptying the trash, backup and restore, export, changing settings, changing the lock. The assistant answers 'ده مش متاح بالصوت'.
- Voice confirmation applies to the latest card only, within the window, with the word lists of 2.4. Other speech is ignored. After the window, the card stays and waits for taps.
- When the app lock is on, the assistant works only after unlocking, and its content is hidden in the recent-apps screen.
- All understanding runs on the phone. The app sends nothing anywhere, has no INTERNET permission, and CI checks that the merged release manifest does not contain it.
- The in-app help explains that speech recognition is done by Android and may use Google speech services depending on the phone.
- The list of unrecognized sentences is text only and stays on the phone.

8. Settings and local data

Settings > المساعد الصوتي:

- Spoken replies (ON by default), speech speed, speak sensitive answers (OFF).
- Voice confirmation (ON), confirmation window in seconds (5).
- Recognition language: Arabic ar-EG by default (English later).
- Keep the list of unrecognized sentences (ON), with 'عرض / View' and 'مسح / Clear'.
- 'اختبار الميكروفون' wizard: records one sentence, shows the transcript, and says whether the phone looks offline (using the existing network-state permission).
- 'اختبار الصوت': speaks a sample.
- A help page with example sentences for every command and question.

Data: one new table unrecognized_phrases (id, text, at, handled, plus the common columns), a few settings keys, and JSON assets for patterns and synonyms (categories, habits, statuses, reasons, month and weekday names). No audio is stored. Later option, not now: owner-defined aliases for habits and categories.

9. Code structure

lib/features/assistant/ with: assistant_sheet, assistant_controller (state machine: idle, listening, processing, card, speaking), speech_input_service, tts_service, text_normalizer, number_words_ar, date_time_resolver, intent_parser (interface), rule_based_intent_parser, patterns and synonyms assets, action_planner, confirmation_card, answer_composer, speech_composer, plural_ar, unrecognized_log. Tests in test/unit/assistant/. Business logic stays in the existing services.

10. Tests

10.1 Phrase corpus. A committed file test/assistant_corpus_ar.json with at least 150 entries: phrase, the fixed now, expected intent and expected slots. It includes every example of this file and, later, the owner's own real sentences. The build fails if the committed corpus does not pass 100 percent.

10.2 Unit tests. Normalizer; number words (including 'ألف وخمسمية' and 'ونص'); DateTimeResolver with every expected result of 4.4 and these times: '5 الساعة' → 17:00, '9 الساعة' → 09:00, '8 بالليل' → 20:00, '5 العصر' → 17:00, 'الضهر' → 12:00, 'ستة وربع' → 18:15; the Arabic plural helper; the confirmation keyword matcher; task, category, habit and reason matching; the ActionPlanner; the CI manifest check.

10.3 Speech recognition test on the owner's phone. The agent prepares a script and a report template. The owner reads 30 sentences: 10 commands with amounts, 10 with dates and times, 5 questions, and 5 with Egyptian dialect words. Run it online and in airplane mode. For each sentence record the transcript, whether numbers and dates were right, and whether the intent was recognized. Write the results and the conclusions in DECISIONS.md. The owner decides whether to continue to step 6.

10.4 UI tests with a fake speech source that injects transcripts, for every card type and for voice confirmation.

10.5 Speech output test. Check whether an Arabic voice exists on the phone, speak the five answer types, and report the quality.

11. Acceptance tests (the owner, on his phone)

1. The assistant opens from the Today microphone button, from More, and from the app shortcut.
2. Say 'سجّل بكرة أكلم العميل الساعة 5'. The card shows a task titled 'أكلم العميل', tomorrow's date in words, 5:00 م, and the default reminder. Confirm by tap: it appears tomorrow in Today with its reminder. Undo works.
3. Say 'انا بكرة عايز اعمل مونتاج الاعلان' (no command word). A task card appears.

4. Say 'سجّل شاي بعشرة جنيه'. The card shows the expense, category شاي وقهوة, 10 EGP, side. Confirm by saying 'تمام'. It is saved and appears in Finance.
5. Say 'كوبايتين قهوة': the coffee habit gets +2. Say 'صليت الفجر': the Fajr item is ticked.
6. Say 'اجّل مهمة التقرير لبكرة لان مفيش فلوس'. The card shows postponed, tomorrow's date and the reason 'مفيش فلوس'. With two tasks called التقرير, the assistant asks which one.
7. Say 'سجّل فكرة صوتية'. The recorder opens, a voice-only idea is saved, and it plays back.
8. Ask 'ورايا ايه النهارده؟'. An answer card appears and is spoken, and it matches the Today screen. Ask 'ورايا ايه بعد يومين؟': it matches the tasks of that date.
9. Ask 'الاسبوع اللي فات عملت ايه؟'. The numbers equal the Stats screen for that week.
10. Ask 'ايه اللي حصل يوم 12/9؟'. The assistant first says the date in words ('12 سبتمبر 2026'), then shows that day.
11. Say a sentence the assistant cannot parse. It answers 'مفهمتش، سمعت: ...'. The button 'افتح الاضافة بالنص ده' opens quick add with the text in the title, and the sentence appears in the unrecognized list.
12. Say 'امسح مهمة التقرير'. The answer is 'ده مش متاح بالصوت'.
13. Finance and health answers are text only by default. Turning on 'قراءة الردود الحساسة صوتياً' makes them spoken. Masked amounts are never spoken.
14. Airplane mode: the result is written in the report of 10.3. The typed-sentence field works in any case. The CI manifest check shows no INTERNET permission.
15. A 20-second sentence with pauses, using tap to start and tap to stop, is captured completely, with no lost words at session boundaries.
16. With the app lock on, the assistant needs the unlock and the recent-apps screen hides it.
17. The new APK installs over the previous one with the data intact.

12. Build steps

Prerequisites: Rounds 0 to 6 of CHANGES.md are done, so the modules exist. Round 7 (timeline) is recommended before this work, because the question 'ايه اللي حصل يوم ...' uses it. If this work starts earlier, that question answers with the limited recap of section 6.

- **Step 1 — Foundations, no UI:** text normalizer, Arabic number words, DateTimeResolver, plural helper, with the tests of 10.2.
- **Step 2 — Parser:** RuleBasedIntentParser. First the commands create_task, create_errand, create_reminder, log_expense, log_habit, create_note, create_idea. Then update_status, log_income, wishlist, health, record_voice_note. Then the questions. The corpus test of 10.1.

- **Step 3 — Cards and execution, typed input only:** ActionPlanner, Confirmation cards, execution through the existing services, Undo, and the assistant sheet with the typed field only (no voice yet). The owner tests by typing the example sentences.
- **Step 4 — Answers:** Answer cards and the SpeechComposer (text only) for all questions, matching the dashboards.
- **Step 5 — Voice input:** tap to start and tap to stop with chained sessions, and the speech recognition report of 10.3. STOP here and report. The owner decides whether to continue.
- **Step 6 — Voice output and settings:** text-to-speech replies, voice confirmation, the assistant settings, the sensitive-answers rule, and the microphone and voice test wizards.
- **Step 7 — Entry points and polish:** the microphone button, the More item, the app shortcut, the voice-note and voice-idea commands, the unrecognized list, the help page, and the documents.
- **Optional later, only if the owner asks:** English commands, owner-defined aliases, navigation commands ('افتح المصروفات'), and a smarter parser plugged in behind the IntentParser interface.

After each step: tests, README, CHANGELOG and DECISIONS, commit, tag, CI APK, a plain-Arabic test checklist, and wait for approval.

13. Risks and open items

- Recognition of the Egyptian dialect, especially numbers and dates, is unknown until the test of 10.3. The cards show everything in words so mistakes are caught before saving.
- The Android recognizer may stop after silence. The chained sessions of 3.2 must be tested. If they prove unreliable, propose alternatives (for example an on-device speech model, which is heavy) and ask the owner before building them.
- Offline recognition depends on the phone's speech packages. Never claim offline support unless it was tested.
- An Arabic voice for speech output may need to be installed on some Xiaomi phones.
- Rule-based coverage is limited. The unrecognized list and the owner's real sentences extend it. Expect several rounds of additions.
- One command per sentence: compound sentences are not split.
- Open items for the owner: send 10 sentences he really says (commands and questions), confirm the default reminder type for voice-created tasks, and confirm the name of the app shortcut.

14. Kickoff messages (the owner pastes one per step)

Start with step 1. Before step 5 starts, wait for the owner's go-ahead.

> ابدأ من VOICE.md فقط. نفّذ الخطوة N فقط (القسم 12). اقرأ القسمين 0 و7 مع أقسام الخطوة. اعمل خطة قصيرة بالعربي وانتظر موافقتي، وبعدها نفّذ. ما تبدأش خطوة تانية، وأي حاجة مش واضحة اسألني فيها. في الآخر: شغّل الاختبارات، حدّث README وCHANGELOG وDECISIONS، اعمل commit وtag باسم voice-N وRelease رسمي للـ APK، وادّيني خطوات التجربة على موبايلي.